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Mechanics of Human Movement
Course Progress 90.2% (11 of 13)
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Week 7: Assignment 7

Due date on 2026-09-09, 23:59 IST

The due date for submitting this assignment has passed. Your last recorded submission was on 2026-09-01, 22:40 IST.

1. Which of the following quantities are required to perform an inverse dynamic analysis? 1 Point

☐ Joint reaction forces between the body segments
☑ Segmental kinematics
☐ Muscle tensions
☑ Segmental parameters such as mass and moment of inertia

Yes, the answer is correct. Score: 1

Accepted answer is Segmental kinematics, Segmental parameters such as mass and moment of inertia

2. The output from an inverse dynamics analysis consists of the following: 1 Point

● Joint reaction forces between the body segments
○ Segmental kinematics
○ Muscle tensions
○ Segmental parameters such as mass and moment of inertia

Yes, the answer is correct. Score: 1

Accepted answer is Joint reaction forces between the body segments

3. For a rigid link undergoing general planar motion, 1 Point

☑ The angular velocity does not change from point to point in the link
☐ The linear velocity is the same for all points in the link
☐ The angular acceleration varies from point to point on the link
☑ The linear acceleration varies from point to point on the link

Yes, the answer is correct. Score: 1

Accepted answer is The angular velocity does not change from point to point in the link, The linear acceleration varies from point to point on the link

4. Which of the following measurement systems would you need to perform inverse dynamics on a specific human subject? 1 Point

☑ Motion Capture System
☑ Force Platforms
☐ ECG
☐ VO_2 Analyser

Yes, the answer is correct. Score: 1

Accepted answer is Motion Capture System, Force Platforms

5. A body is more likely to remain in balance even if disturbed, if its 1 Point

☐ Centre of mass lies outside base of support
☐ Centre of mass lies on the base of support
☑ Centre of mass remains within the base of support
☐ Centre of mass is at a low elevation with respect to the ground

Partially Correct. Score: 0.5

Accepted answer is Centre of mass remains within the base of support, Centre of mass is at a low elevation with respect to the ground

6. Muscle tension calculated from the net joint moment when a single muscle is assumed to be acting about a joint 1 Point

● Is likely to be less than the actual muscle tension
○ Is an accurate estimate of the actual muscle tension
○ Is likely to be an overestimate of the actual muscle tension
○ Is unrelated to the net joint moment

Yes, the answer is correct. Score: 1

Accepted answer is Is likely to be less than the actual muscle tension

7. While drawing the free body diagram of a system the following force(s) is/are NOT considered 1 Point

○ Weight of the system
● Forces exerted by the system
○ Forces exerted at the boundaries on the system
○ External forces on the system

Yes, the answer is correct. Score: 1

Accepted answer is Forces exerted by the system

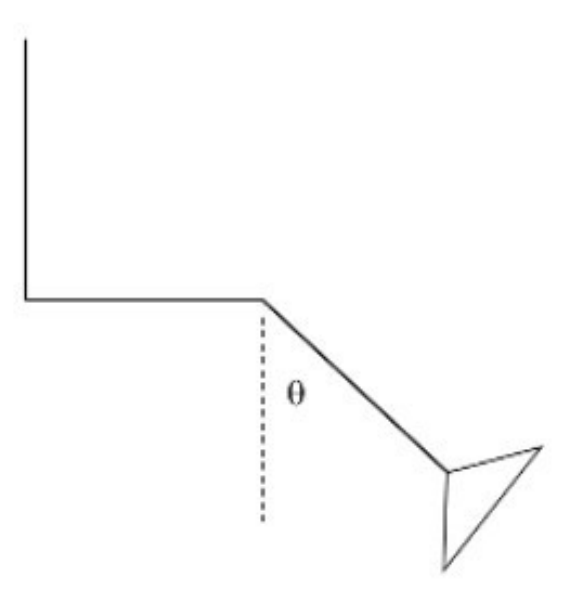
8. Select the major agonist, antagonist pair that helps a gymnast in maintaining a straight body posture while swinging down from a bar. 1 Point

● Back muscles, Abdominal muscles
○ Abdominal muscles, Back muscles
○ Abdominal muscles, Shoulder Muscles
○ Shoulder muscles, Abdominal Muscles

No, the answer is incorrect. Score: 0

Accepted answer is Abdominal muscles, Back muscles

Based on the data, Answer from Questions 9-15



Let θ be the angle made by the shank with the vertical axis. At an instant when θ = 45 degrees, ω = 5 rad/s and α = 100 rad/s², and assuming the weight of the shank and foot to be 50 N and ankle weights of 10 N, moment arm of the quadriceps muscle about the knee to be 4 cm, location of the COM of the shank and foot to be 22 cm from the knee joint, location of the ankle at 50 cm from the knee, the line of action of the quads muscle to be at 24 degrees to the axis of the lower leg and parallel to the thigh, and the moment of inertia of the shank+foot about the knee joint is 0.25 kgm², answer the following (Assume g=10m/s²):

A person is doing leg lifts using ankle weights to strengthen his quadriceps muscles.

9. The new COM of the lower leg is located at _____ from the knee. 1 Point

○ 22.7 cm
○ 25.7 cm
● 26.7 cm
○ 50 cm

Yes, the answer is correct. Score: 1

Accepted answer is 26.7 cm

10. The MOI of the system about the knee is 1 Point

○ 0.25 kgm^2
○ 0.35 kgm^2
● 0.5 kgm^2
○ 0.65 kgm^2

Yes, the answer is correct. Score: 1

Accepted answer is 0.5 kgm^2

11. The net torque produced about the knee joint is _________ 1 Point

○ 25 Nm
○ 50 Nm
● 60 Nm
○ 70 Nm

No, the answer is incorrect. Score: 0

Accepted answer is 50 Nm

12. The tension in the patellar tendon is ___________ 1 Point

○ 488 N
● 1533 N
○ 1307 N
○ 3423 N

Yes, the answer is correct. Score: 1

Accepted answer is 1533 N

13. The moment of inertia of the system about its CoM is ___________ kgm^2 1 Point

○ 0.007
● 0.07
○ 0.7
○ 7

Yes, the answer is correct. Score: 1

Accepted answer is 0.07

14. The reaction force at the tibiofemoral joint is ___________ N 1 Point

○ 800
○ 684
● 1384
○ 925

Yes, the answer is correct. Score: 1

Accepted answer is 1384

15. What is the magnitude of the linear acceleration of the ankle at this instant? 1 Point

○ $25.75m/s^2$
● $51.5m/s^2$
○ $31.5m/s^2$
○ $57m/s^2$

Yes, the answer is correct. Score: 1

Accepted answer is $51.5m/s^2$